Testcases (examples/main) included in the full DualSPHysics package v5.4 in http://dual.sphysics.org/index.php/downloads/

<table>
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<td>

01_DamBreak

- 3-D dam break flow impacting on a structure: numerical velocity, pressure and force are computed. Video
- 2-D dam break and validation data from [Koshizula and Oka, 1996] experiment. Video

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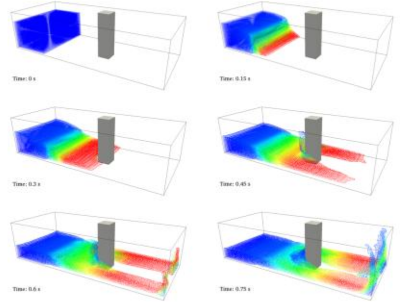
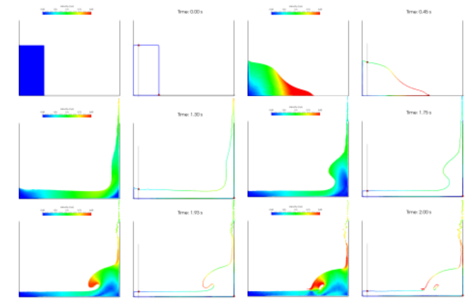
<td></td>
<td></td>
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02_Periodicity

- 2-D case with Periodicity in X direction. Video
  Density diffusion term is applied.

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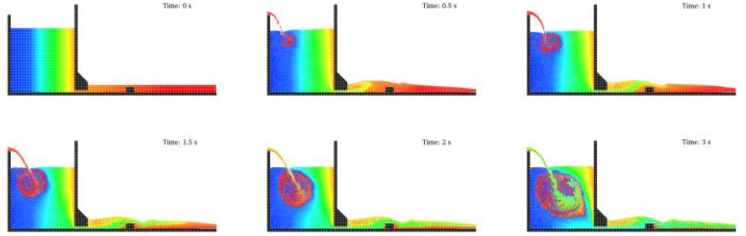
<td colspan="2"></td>
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03_MovingSquare

- 2-D case with square that moves with rectilinear movement. Video
  Example with no gravity; parameter “b” needs to be specified by the user.
  Shifting is used for this internal flow (no need to detect free surface).

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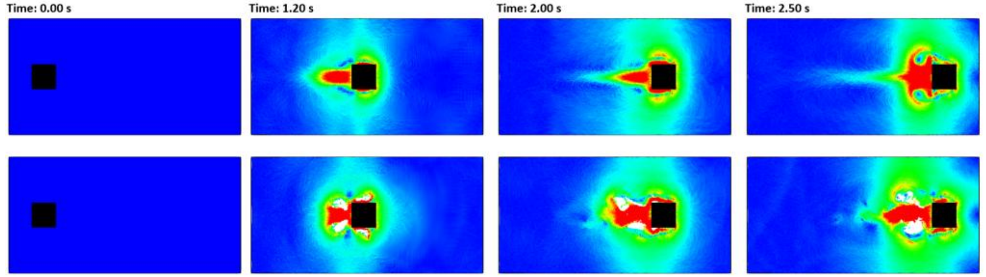
<td colspan="2">
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04_ExternalForces

- External acceleration is loaded from a file and applied to two different volumes of fluid. Video
  Density diffusion term is applied.

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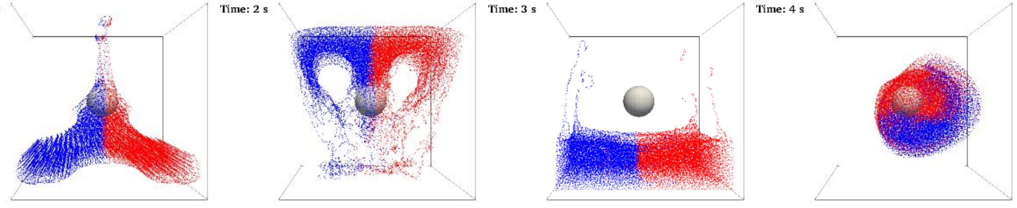
<td colspan="2">
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05_SloshingTank

- 2-D sloshing tank that reads the rotational movement of the tank itself from a file. Video
- 2-D sloshing tank that reads external acceleration from a file.
  Validation with SPHERIC Benchmark #10 where pressure is computed.
  Density diffusion term is applied.

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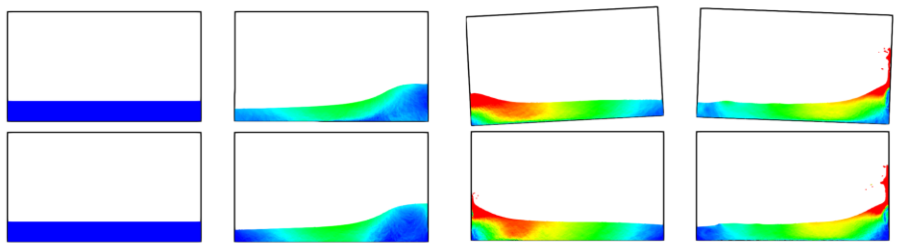
<td colspan="2">
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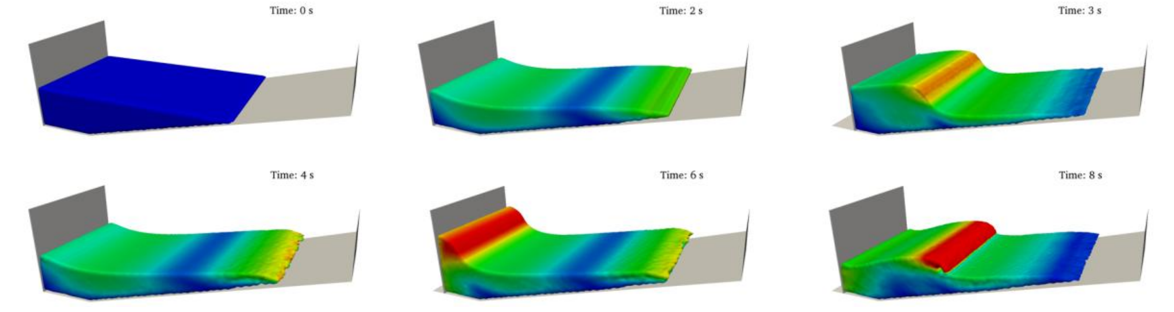
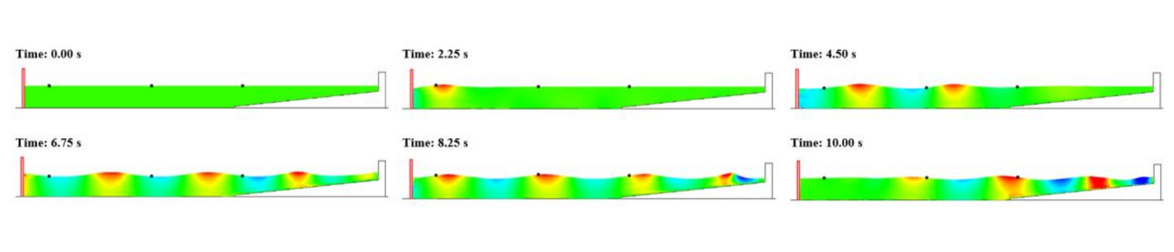
06_WAVEMAKER

- 3-D tank with Periodicity in Y direction and piston with sinusoidal movement. Video
  Density diffusion term is applied.



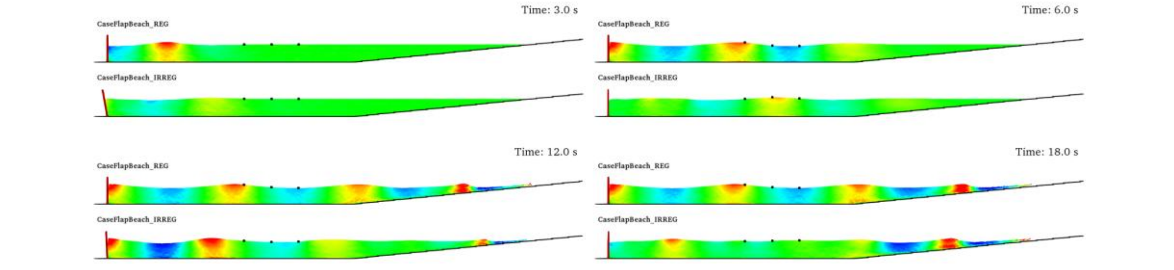
07_WAVEMAKERFILE

- 2-D tank with piston motion loaded from external file and external structure (STL). Video
  Validation data from CIEMito experiment: numerical computation of wave surface elevation and force exerted onto the wall.



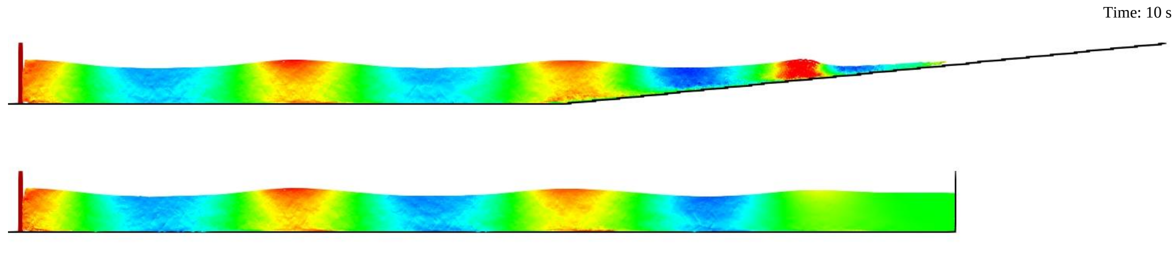
08_WAVESFLAP

- 2-D regular waves generated with flap and comparison with 2nd order wave theory (beach). Video
- 2-D irregular waves generated with flap and comparison with 2nd order wave theory (beach).



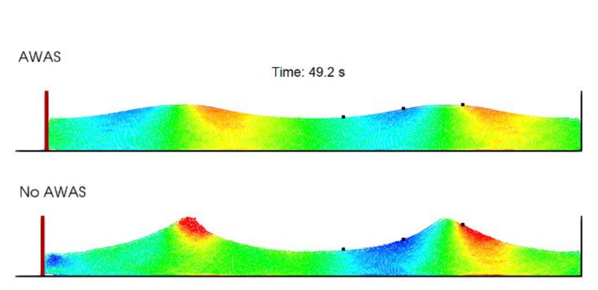
09_WAVESPISTON

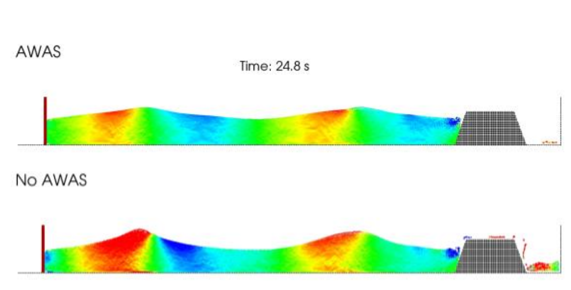
- 2-D regular waves with piston and comparison with 2nd order wave theory (beach & damping). Video
- 2-D irregular waves with piston and comparison with 2nd order wave theory (beach & damping). Video



10_WAVESPISTONAWAS

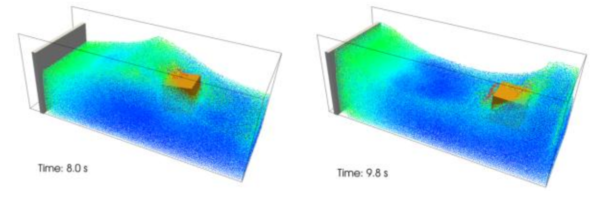
- 2-D regular waves generated with piston interacting with a vertical wall with and without AWAS. Forces against the wall and dike with and without AWAS are compared. Video
- 2-D regular waves generated with piston interacting with a trapezoidal dike with and without AWAS. Forces against the wall and dike with and without AWAS are compared. Video

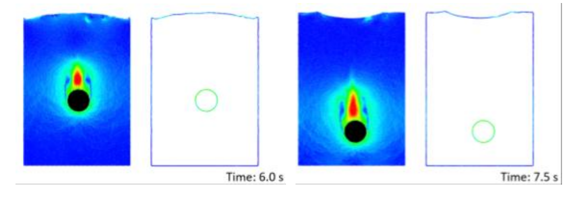




11_Floating

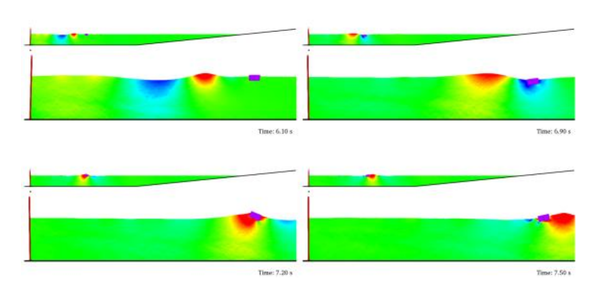
- 3-D floating box in a wave tank with Periodicity in Y direction and piston with sinusoidal movement. Delta-SPH is used. [Video]
- 2-D falling sphere that uses laminar+SPS viscosity. Validation data from [Fekken, 2004] and [Moyo and Greenhow, 2000]. [Video]

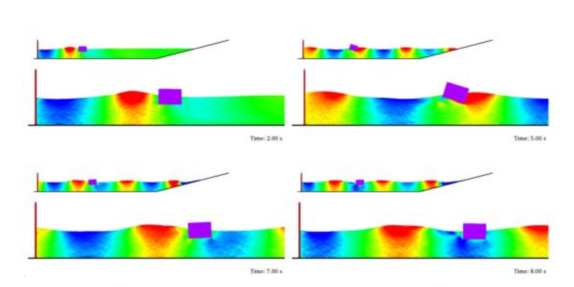




12_FloatingWaves

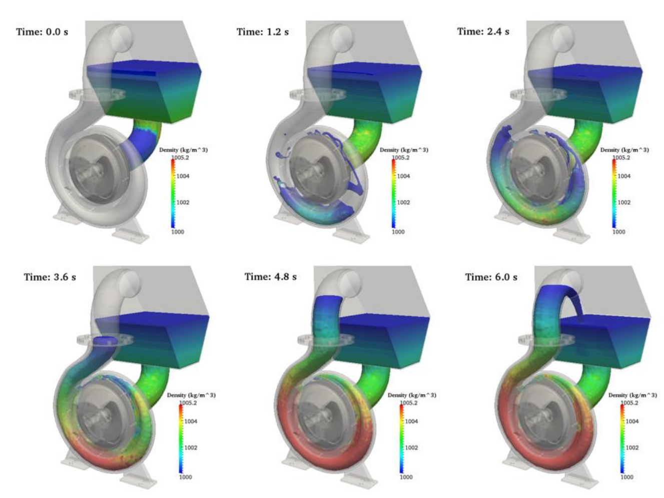
- 2-D floating box under the action of non-linear waves with flap that reads rotational motion from a file and uses laminar+SPS viscosity. [Video] Validation data (motions of the box) from [Hadzic et al., 2005].
- 2-D floating box under the action of regular waves with piston. [Video] Validation data (motions of the box) from [Ren et al., 2015].





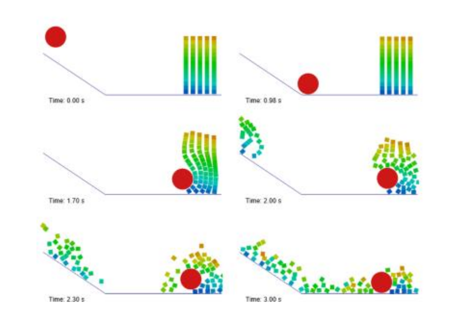
13_Pump

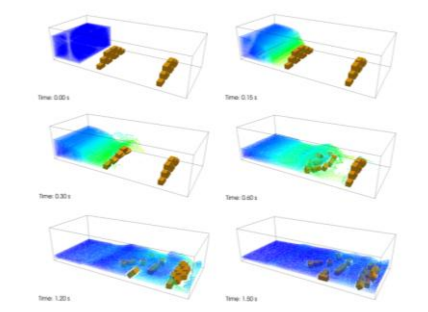
- 3-D external geometries are imported (STL) and filling algorithm is used. Rotational movement is imposed. [Video]

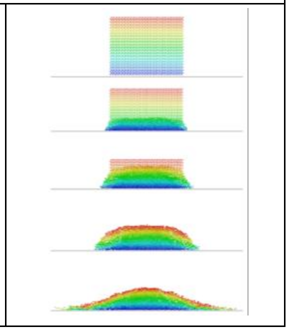


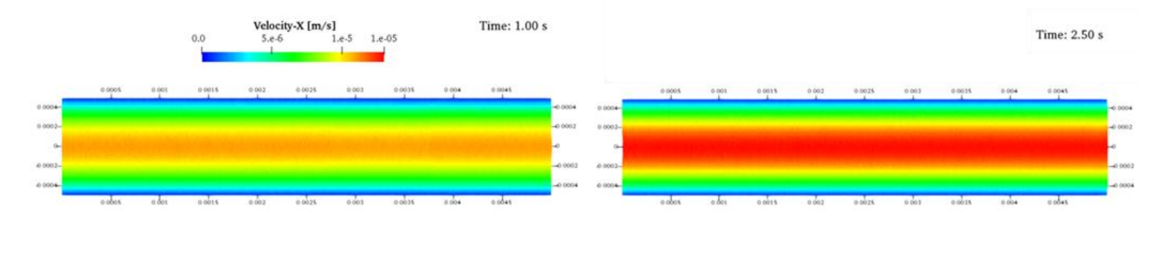
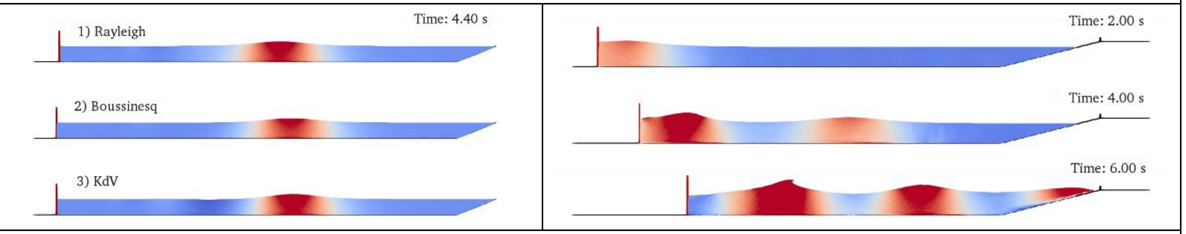
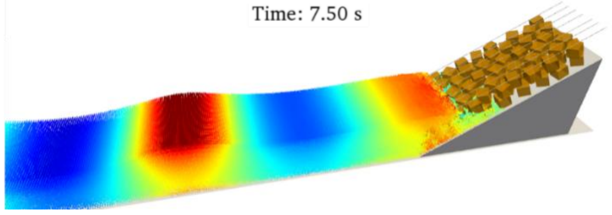
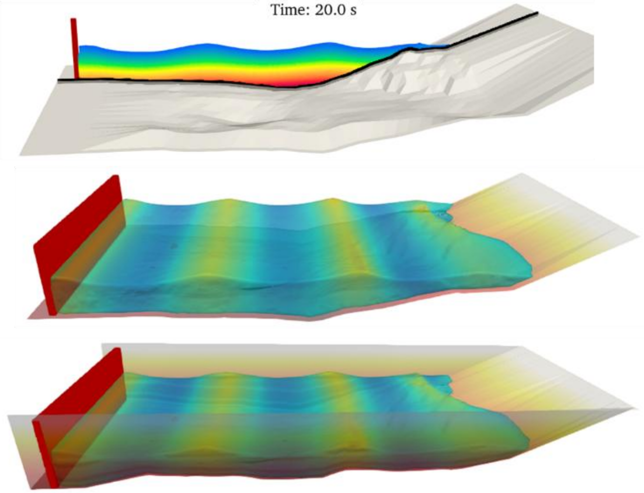
14_DEM

- 2-D case only with DEM of a ball that impacts with blocks. Example without fluid particles. [Video]
- 3-D dam-break and blocks where interaction between blocks and with walls used DEM and properties of materials. [Video]
- 2-D case with 2000 floating objects that interact in terms of DEM approach. [Video]



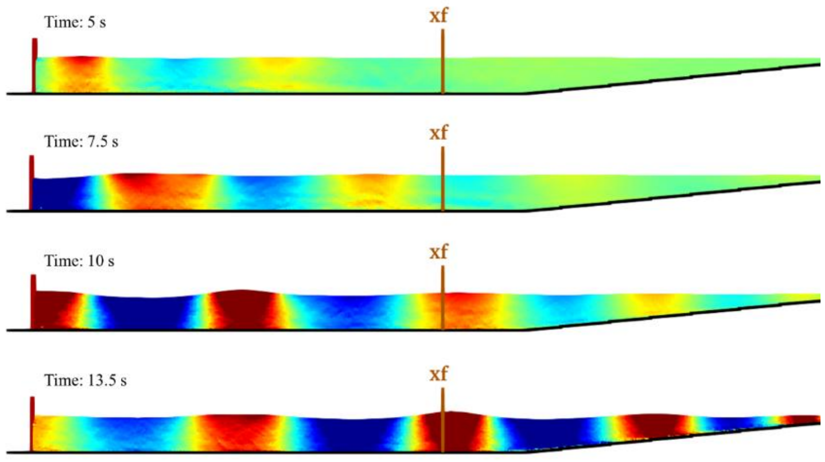




15_POISEUILLE • 2-D case of Poiseuille flow with laminar+SPS viscosity and using high resolution. Video	
16_SOLITARYWAVES • 2-D solitary wave generated with 3 different theories. Video • 2-D case of triple solitary waves. Video	
17_WAVERUNUP • 3-D regular waves interacting with a layer armour breakwater (STL) • Gauge system is used to compute Run-up. Video Validation data from CIEMito: wave surface elevation and wave run-up.	
18_BATHYMETRY • 2-D simulation of regular waves using bathymetry automatically generated from XYZ points. Video • 3-D bathymetry automatically generated starting from XYZ points for open periodic conditions. • 3-D bathymetry automatically generated starting from XYZ points for closed domain. Video	

19_FocusedWaves

- 2-D focused waves are generated with piston following the NewWave theory [Whittaker et al. 2017].
  Video



20_Damping

- 2-D wave absorption using numerical damping. The simplest case.
  Video
- 3-D simulation with numerical damping in the form of an angle.
  Video
- Numerical damping for box-shaped tanks.
  Video
- Numerical damping for cylinder-shaped tanks.
  Video